HBL Engineering Limited

Rail Signaling Group

In-Process QC Checklist for Stationary Kavach Installation

QC Engineer Name: sushma v1.2

Station ID	Station Name	Zone	Division	Section Name	Initial Date	Updated Date
25285282	Station	ER	Howrah-COO	Howrah-COO	23/03/2026	23/03/2026

Installation Status	☑ Completed	☐ Not Completed

Total Points	Closed Points	Open Points
154	154	0

1.0 Serial Number Verification

S.No	Units	Status	Observation	Image
1.1	Stationary Kavach Unit	Matching		
1.2	Peripheral Processing Card 1	Matching		
1.3	Peripheral Processing Card 2	Matching		
1.4	Vital Computer Card -1	Matching		
1.5	Vital Computer Card -2	Matching		
1.6	Vital Computer Card -3	Matching		
1.7	Voter Card -1	Matching		
1.8	Voter Card -2	Matching		
1.9	Vital Gateway Card 1	Matching		
1.10	Vital Gateway Card 2	Matching		
1.11	Vital Gateway Card 3 (NMS)	Matching		
1.12	EI Gateway-1	Matching		
1.13	EI Gateway-2	Matching		
1.14	Field Scanner Card 1	Matching		
1.15	Field Scanner Card 2	Matching		
1.16	Field Scanner Card 3	Matching		
1.17	Field Scanner Card 4	Matching		
1.18	Field Scanner Card 5	Matching		
1.19	Field Scanner Card 6	Matching		
1.20	Field Scanner Card 7	Matching		

S.No	Units	Status	Observation	Image
1.21	Field Scanner Card 8	Matching		
1.22	RIU communication card 1-Host	Matching		
1.23	RIU communication card 2-Host	Matching		
1.24	RS 232-OFC converter 1 (STATION)	Matching		
1.25	RS 232-OFC converter 2 (STATION)	Matching		
1.26	RS 485-OFC converter (STATION)	Matching		
1.27	FIU Termination Card 1	Matching		
1.28	FIU Termination Card 2	Matching		
1.29	FIU Termination Card 3	Matching		
1.30	FIU Termination Card 4	Matching		
1.31	FIU Termination Card 5	Matching		
1.32	FIU Termination Card 6	Matching		
1.33	FIU Termination Card 7	Matching		
1.34	FIU Termination Card 8	Matching		
1.35	DPS Card 1	Matching		
1.36	DPS Card 2	Matching		
1.37	GPS & GSM Antenna-1	Matching		
1.38	GPS & GSM Antenna-2	Matching		
1.39	SMOCIP Unit	Matching		
1.40	SMOCIP Termination Panel	Matching		
1.41	Station Termination Panel	Matching		
1.42	Stationary PDU Box	Matching		
1.43	IPS PDU Box	Matching		
1.44	DC-DC Converter (Relay racks)	Matching		
1.45	RTU-1	Matching		
1.46	RTU-2	Matching		
1.47	Station Radio Power Supply card-1	Matching		
1.48	Station Radio Power Supply card-2	Matching		
1.49	Next Gen/. Cal Amp Radio Modem-1	Matching		
1.50	Next Gen/. Cal Amp Radio Modem-2	Matching		
1.51	RS 232-OFC converter 1 (RTU-1)	Matching		
1.52	RS 232-OFC converter 2 (RTU-2)	Matching		

S.No	Units	Status	Observation	Image
1.53	RS 485-OFC converter (SM-OCIP)	Matching		
1.54	RIU	Matching		
1.55	RIU Power Supply Card-1	Matching		
1.56	RIU Power Supply Card-2	Matching		
1.57	RIU communication card 1-Remote	Matching		
1.58	RIU communication card 2-Remote	Matching		
1.59	Field Scanner Card 1	Matching		
1.60	Field Scanner Card 2	Matching		
1.61	Field Scanner Card 3	Matching		
1.62	Field Scanner Card 4	Matching		
1.63	RIU Battery Charge Cum Filter-1	Matching		
1.64	RIU Battery Charge Cum Filter-2	Matching		
1.65	relay rack	Matching		

2.0 Building (New Building Constructed by HBL)	**Not Applicable**

3.0 Tower (New Tower Constructed by HBL)	**Not Applicable**

4.0 Power Supply

S.No	Aspect	Requirement	Status	Observation	Image
		4.1 PDU(Power Distribution Unit)			
4.1	Installation	IPS PDU unit shall be mounted on the wall using M8 insulators, secured with M8 bolts, and tightened to a torque of 20 Nm as per diagram 5 16 76 0053	Yes		
4.1.1	Installation	Station PDU unit shall be mounted on the wall using M8 insulators, secured with M8 bolts, and tightened to a torque of 20 Nm as per diagram 5 16 76 0054	Yes		
4.1.2	Cabling	All external cables entering the PDU shall pass through cable glands and ensure no cable entry opening shall be used without a cable gland.	Yes		

S.No	Aspect	Requirement	Status	Observation	Image
4.1.3	Cabling	Output connections shall be maintained as per the Station PDU schematic diagram 5 16 49 0671	Yes		
4.1.4	Cabling	Ensure lugs with sleeves / Ferrules are properly crimped and inserted into the terminal; no loose strands shall be left.	Yes		
4.1.5	Earthing	PDU units shall be connected to the ring earth/busbar using a 10 sq.mm green/yellow earthing wire, and bolts shall be tightened to a torque of 8 Nm with torque marking applied.	Yes		
4.1.6	Earthing	Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.	Yes		
4.1.7	Functionality	Functional testing shall be performed as per the PDU test procedure 5 53 69 0001.	Yes		
4.2 DC-DC Converter					
4.2	Installation	DC-DC Converter unit shall be installed as per the approved floor plan drawing, mounted on floor using M10 insulators, secured with M10 bolts, and tightened to a torque of 40 Nm as per diagram 5 16 76 0055	Yes		
4.2.1	Cabling	All external cables entering the unit shall pass through cable glands and ensure no cable entry opening shall be used without a cable gland.	Yes		
4.2.2	Earthing	Unit shall be connected to the ring earth/busbar using a 10 sq.mm green/yellow earthing wire, and bolts shall be tightened to a torque of 8 Nm with torque marking applied.	Yes		
4.2.3	Earthing	Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.	Yes		
4.2.4	Functionality	DC-DC converter output voltage shall be minimum 24 V DC, +/- 5% (22.8 VDC to 25.2 VDC)	Yes		

5.0 Kavach Equipment

S.No	Aspect	Requirement	Status	Observation	Image
5.1 Kavach Unit					
5.1	Installation	Kavach unit shall be installed as per the approved floor plan drawing, mounted on the floor using M10 insulators, secured with M10 bolts, and tightened to a torque of 40 Nm as per diagram 5 16 76 0056.	Ok		
5.1.1	Cabling	All external cables shall enter through cable glands only. No unused cable entries left open. Ensure mill connector shall be locked properly.	Ok		
5.1.2	Earthing	Unit shall be connected to the ring earth/busbar using a 10 sq.mm green/yellow earthing wire, and bolts shall be tightened to a torque of 8 Nm with torque marking applied.	Ok		
5.1.3	Earthing	Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.	Ok		
5.1.4	Earthing	Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.	Ok		
5.1.5	Termination Unit	Station Kavach Termination Unit shall be wall-mounted near the Kavach unit using insulators, secured with M8 bolts, and tightened to a torque of 20 Nm as per diagram 5 16 76 0045.	Ok		
5.1.6	Termination Unit	OFC cables for SMOCIP and RTU shall be spliced as per diagram 5 16 49 0559, and proper bunching and routing shall be ensured.	Ok		
5.2 SMOCIP(Station Master's Operation-Cum-Indication Panel)					
5.2	Installation	SMOCIP shall be installed in the Station Master’s room at an ergonomic height. The panel shall be securely fixed using M6 screws and tightened to a torque of 8 Nm with torque marking applied, as per diagram 5 16 76 0040	Ok		

S.No	Aspect	Requirement	Status	Observation	Image
5.2.1	Termination Unit	SMOCIP Termination Unit shall be wall-mounted near the SMOCIP unit, using insulators, secured with M6 bolts, and tightened to a torque of 8 Nm as per diagram 5 16 76 0046	Ok		
5.2.2	Termination Unit	Power and OFC cables from Kavach termination unit shall be terminated as per diagram 5 16 49 0559.	Ok		
5.2.3	Termination Unit	OFC cables of SMOCIP shall be spliced as per diagram 5 16 49 0559, and proper bunching and routing shall be ensured.	Ok		
5.2.4	Earthing	Unit shall be connected to the ring earth/busbar using a 10 sq.mm green/yellow earthing wire, and bolts shall be tightened to a torque of 8 Nm with torque marking applied.	Ok		
5.2.5	Earthing	Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.	Ok		
5.2.6	Functionality	System Health LED shall blink and ensure SYSTEM OK along with the respective station name shall be displayed on the Display.	Ok		
5.2.7	Functionality	Verify that pressing the SOS and Common buttons on the SM-OCIP increments the mechanical counter by one.	Ok		
5.2.8	Checksum	Verify the checksums as per the FAT certificate.	Matching		
5.3 GPS/GSM Antennas					
5.3	Installation	Two antennas shall be installed on the Kavach room rooftop with a minimum separation of 3 meters, grouting shall be carried out as per diagram 5 16 67 0039, and torque of 10 Nm shall be applied for M6 fasteners with torque marking provided.	Ok		
5.3.1	Installation	No obstruction above antennas like tree branches, sun-shades, and ensure open to sky etc.	Ok		

S.No	Aspect	Requirement	Status	Observation	Image
5.3.2	Cabling	Antenna cables shall be routed via diverse paths.	Not Applicable		
5.3.3	Cabling	Separate conduits shall be used and Roof conduits shall be sealed against dust, water, and insects.	Ok		
5.3.4	Cabling	In each antenna, the GPS and GSM cables shall be connected to their respective connectors as per the labels provided on the antenna.	Ok		
5.4 RIU(Remote Interface Unit)					
5.4	Installation	RIU shall be installed on floor using M10 insulators, secured with M10 bolts, and tightened to a torque of 40 Nm as per diagram 5 16 76 0057.	Ok		
5.4.1	Cabling	All external cables entering into the RIU unit shall pass through cable glands and ensure no cable entry opening shall be used without a cable gland.	Not Applicable		
5.4.2	Cabling	OFC patch cords shall be properly tagged to identify default and standby links.	Ok		
5.4.3	Earthing	Unit shall be connected to the ring earth/busbar using a 10 sq.mm green/yellow earthing wire, and bolts shall be tightened to a torque of 8 Nm with torque marking applied.	Ok		
5.4.4	Earthing	Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.	Ok		
5.4.5	FDMS Box installation	FDMS Box shall be installed in the 15U/17U rack of the RIU with proper wall mounting, and OFC splicing shall be carried out as per the network drawing.	Ok		

6.0 RF Communication Equipment On Tower

S.No	Aspect	Requirement	Status	Observation	Image
6.1 RTU(Radio Tower Unit)					
6.1	RTU Fixing	Both RTUs shall be firmly secured to the tower platform using M12 bolts and nuts, and a torque of 85 Nm shall be applied as per diagram 5 16 67 0983.	Ok		
6.1.1	RTU Fixing	Ensure RTU doors are fully closed and locked.	Ok		
6.1.2	RTU Earthing	RTU shall be properly earthed by connecting a 35 sq.mm green to GI strip earthing conductor from the RTU earthing bolt to the designated earth pit-4, as per diagram 5 16 76 0043.	Ok		
6.1.3	RTU Earthing	Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.	Ok		
6.1.4	OFC cable termination	OFC cable from the Relay Room shall be spliced and terminated in the splice holder inside the RTU. (Ref. Drawing: 5 16 49 0559)	Ok		
6.1.5	OFC cable termination	OFC cables for RTU shall be spliced as per diagram 5 16 49 0559 and ensure bunching and routing shall be done properly.	Ok		
6.1.6	110V Power cable termination	Cable glands used for 110 V DC power cable entry into RTU shall be firmly tightened.	Ok		
6.1.7	110V Power cable termination	110 V DC power cables shall be terminated inside RTU as per approved drawing 5 16 49 0672.	Ok		
6.1.8	110V Power cable termination	Ensure lugs with sleeves / Ferrules are properly crimped and inserted into the terminal; no loose strands shall be left.	Ok		
6.1.9	Cabiling	LMR 600 connection with proper routing; and clamping; No Joints shall be done, as per Tower SOP 5 16 90 0018.	Ok		
6.2 RF Antenna Installation					
6.2	RF antenna installation and Audit	RF antenna installation and orientation shall be done as per 10.2dBi omni-directional anteena diagram 5 16 67 0983.	Ok		

S.No	Aspect	Requirement	Status	Observation	Image
6.2.1	RF antenna installation and Audit	RF antenna installation audit report from the installation contractor shall be available in WFMS and there shall be no open points in the audit report.	Not Applicable		

7.0 OFC Networking Rack

S.No	Aspect	Requirement	Status	Observation	Image
7.1	Installation	Network rack shall be installed as per approved floor plan and installation shall be done as per diagram 5 16 76 0058.	Ok		
7.1.1	Installation	Patch cord routing shall be neat & bend radius to be maintained.	Not Applicable		
7.2	Cabling	All external cables entering the network rack shall pass through the grommets.	Ok		
7.2.1	Cabling	OFC cables shall be marked using naming tie-tags for easy identification of default and standby links.	Ok		
7.3	Earthing	All networking modules inside the rack shall be connected to the rack chassis using 2.5 sq.mm green/yellow earthing wire.	Ok		
7.3.1	Earthing	Network rack shall be connected to ring earth using a 10 sq.mm green/yellow earthing wire.	Ok		
7.3.2	Earthing	Bolts shall be tightened to a torque of 8 Nm, and torque marking shall be applied using yellow paint.	Ok		
7.4	FDMS Box Installation in Network Rack	OFC cables shall be spliced in the FDMS box as per network drawing.	Ok		
7.4.1	FDMS Box Installation in Network Rack	FDMS boxes shall be clearly marked to identify up and down links.	Ok		
7.5	OFC cable continuity check, after splicing	OTDR test reports shall be available.	Ok		

8.0 Relay Rack

S.No	Aspect	Requirement	Status	Observation	Image
8.1	Installation	Relay rack shall be installed as per the approved Floor Plan drawing, mounted on insulators, and secured to the floor by grouting as per diagram 5 16 76 0059.	Ok		
8.2	Wiring	Labeling sleeve shall be used to identify wiring with rack number, row number, relay number & contact type, Labelling shall be provided to relay contact wires at FTC PCBA of Station Kavach.	Not Applicable		
8.2.1	Wiring	For EI Stations, verify all connections as per the approved EI Interface diagrams (vendor specific).	Ok		
8.2.2	Wiring	WAGO terminal details shall be as per interface circuit diagram 5 16 49 0685.	Ok		
8.3	Continuity Test / Bell Test	Completed Station Analyser and Bell Test reports shall be available.	Ok		
8.4	Earthing	Unit shall be connected to the ring earth/busbar using a 10 sq.mm green/yellow earthing wire, and bolts shall be tightened to a torque of 8 Nm with torque marking applied.	Ok		
8.4.1	Earthing	Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.	Ok		

9.0 Earthing

S.No	Aspect	Requirement	Status	Observation	Image
9.1	Installation	Earthing shall be done as per RDSO Spec RDSO/SPN/197/2008 and earth pit placement shall be done as per diagram 5 16 76 0043 .	Ok		
9.2	Earth Resistance Measurement	Earth resistance shall be measured using a calibrated earth resistance tester. The measured value sha Ohm for Stationary Kavach equipment earth pit.	Ok		
9.3	Earth Resistance Measurement	Check the value of earth resistance at earth of the radio tower.	Ok		
9.4	Test Reports	Earth resistance test reports shall be available.	Available		
9.5	Labeling	All earth pits, earthing conductors, and earth points shall be clearly labelled and identifiable.	Ok		

10.0 Indoor Wiring / Cabling

S.No	Aspect	Requirement	Status	Observation	Image
10.1	Redundant cabling	Cabling shall be done as per station connectivity diagram 5 16 49 0614 Segregation of power & communication cables.	Ok		
10.2	SMOCIP	12 Core Signaling cable shall be used for button, counter & power supply. OFC armoured cables shall be used for communication.	Ok		
10.3	No Joints	No joints permitted except inside junction boxes or panels.	Ok		
10.4	Color Coding	Colour codes shall be followed for phase, neutral, and earth.	Ok		
10.5	Tagging/Labeling	Cables shall be tagged at both ends.	Ok		
10.6	Termination & C onnection	Proper lugs/ferrules shall be used and terminals tightened correctly.	Ok		

11.0 Outdoor Cabling

S.No	Aspect	Requirement	Status	Observation	Image
11.1	Cable route plan for OFC & Power cable (Relay Room to Tower)	Railway-approved outdoor signalling cable route plan shall be available.	Ok		
11.2	Cable route markers	Route markers shall be installed and clearly visible.	Ok		
11.3	Cable route markers	Where contractually required, underground RFID markers shall be installed and verified.	Ok		

12.0 RFID Tags

S.No	Aspect	Requirement	Status	Observation	Image
12.1	RFID Tag Installation	Ensure the main RFID tag is installed in the direction specified in the RFID tag layout, and a duplicate RFID tag shall be installed at a distance of 3–5 meters from the main tag, except for turnout tags.	Ok		
12.2	RFID Tag Installation	RFID tag mounting shall be firm and secure, ensuring no gap or mechanical play as per RFID tag installation procedure 5 53 76 0055.	Ok		
12.3	RFID Tag data Validation and Placement verification	System-generated RFID data validation report and Placement verification report shall be available.	Available		

13.0 Components Inspection

S.No	Aspect	Requirement	Status	Observation	Image
13.1	Materials Inspection (Which are not Inspected by RDSO or Consignee)	Verify that all subcontractors and HBL personnel use only approved make and part numbers as per the List of Acceptable I&C Materials EG-EN-FT-34.	Ok		

Note : This QC report is generated by the system and does not need a physical signature.